

1. MATRIX ALGEBRA

$\rho(AB) \leq \min\{\rho A, \rho B\}$; $\rho(A+B) \leq \rho A + \rho B$; $\rho(A) = \rho(A^T) = \rho(A^T A) = \rho(AA^T)$; $\mathcal{C}(A^T) = \mathcal{C}(A^T A)$; $\mathcal{N}(A) = \mathcal{N}(A^T A)$; $\mathcal{C}(A^T) = \mathcal{N}(A)^\perp$; $\text{tr}(AB) = \text{tr}(BA)$; $\text{tr} A = \sum \lambda_i$; $Ax=b$ solvable $\iff b \in \mathcal{C}(A)$; spectral $A=P\Lambda P^T$, $PP^T=P^T P=I$.
 $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ac-bd} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$; $\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1}+A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{bmatrix}$,
 $S=D-CA^{-1}B$; $(A+uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1+v^T A^{-1}u}$.

2. PROJECTIONS [24 Q1, 25 Q4]

Projection TFAE: $\mathbf{P}^2=\mathbf{P}$; $\mathcal{N}(\mathbf{P})=\mathcal{C}(I-\mathbf{P})$; $\rho(\mathbf{P})+\rho(I-\mathbf{P})=n$. **Orthogonal TFAE:** $\mathbf{P}=\mathbf{P}^T\mathbf{P}$; $\mathbf{P}^T=\mathbf{P}$ and $\mathbf{P}^2=\mathbf{P}$. Idempotent $\Rightarrow \lambda \in \{0, 1\}$

$$\text{tr}(\mathbf{P}) = \rho(\mathbf{P})$$

$$\mathcal{C}(\mathbf{P})=S, \mathcal{C}(I-\mathbf{P})=S^\perp, \mathbf{Y}^T\mathbf{P}\mathbf{Y}=\|\mathbf{P}\mathbf{Y}\|^2 \geq 0.$$

$$\mathbf{P}\mathbf{X} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-}\mathbf{X}^T$$

$\mathbf{P}\mathbf{X}\mathbf{X}=\mathbf{X}$; $(I-\mathbf{P})\mathbf{X}=0$; $\rho(\mathbf{P}\mathbf{X})=\rho(\mathbf{X})$; invariant to $(\mathbf{X}^T\mathbf{X})^{-}$; $\mathcal{C}(\mathbf{Z}) \subseteq \mathcal{C}(\mathbf{X}) \Rightarrow \mathbf{P}\mathbf{X}\mathbf{P}\mathbf{Z}=\mathbf{P}\mathbf{Z}\mathbf{P}\mathbf{X}=\mathbf{P}\mathbf{Z}$.

$$\mathbf{X} = [\mathbf{Z} : \mathbf{W}] \Rightarrow \mathbf{P}\mathbf{X} = \mathbf{P}\mathbf{Z} + \mathbf{P}\mathbf{W}|_{\mathbf{Z}}$$

$\mathbf{P}\mathbf{W}|_{\mathbf{Z}}=M(M^T M)^{-1}M^T$, $M=(I-\mathbf{P}\mathbf{Z})\mathbf{W}$, $\rho=\rho(\mathbf{X})-\rho(\mathbf{Z})$.

[24 Q1] $\mathbf{P}, \mathbf{Q}$ symm. orth. proj.: $\mathbf{P}+\mathbf{Q}$ orth. proj. $\iff \mathbf{P}\mathbf{Q}=\mathbf{O}$; then $\rho(\mathbf{P})+\rho(\mathbf{Q})=\rho(\mathbf{P}+\mathbf{Q})$.

$\Leftarrow$: $\mathbf{P}\mathbf{Q}=0 \Rightarrow \mathbf{Q}\mathbf{P}=(\mathbf{P}\mathbf{Q})^T=0 \Rightarrow (\mathbf{P}+\mathbf{Q})^2=\mathbf{P}+\mathbf{Q}$. $\Rightarrow$: $\mathbf{P}\mathbf{Q}+\mathbf{Q}\mathbf{P}=0$; $\times \mathbf{P}$ left, $\times \mathbf{P}$ right, subtract $\Rightarrow \mathbf{P}\mathbf{Q}=\mathbf{Q}\mathbf{P} \Rightarrow 2\mathbf{P}\mathbf{Q}=0$. Rank: $\rho(\mathbf{P}+\mathbf{Q})=\text{tr}\mathbf{P}+\text{tr}\mathbf{Q}$.

[25 Q4] A, B symm., $B^2=B$, $A \succeq 0$, $I-A-B \succeq 0 \Rightarrow AB=BA=0$.

$y \in \mathcal{C}(B) \Rightarrow By=y$; $0 \leq y^T(I-A-B)y = -y^T Ay \Rightarrow y^T Ay=0$; $A=L^T L \Rightarrow \|Ly\|^2=0 \Rightarrow Ay=0$.

3. GENERALISED INVERSES [24 Q3]

$$AGA = A$$

$y \in \mathcal{C}(A) \Rightarrow A^-y$ solves $Ax=y$; all solutions $A^-y+(I-A^-A)z$. A symm.: $A^- = P\Delta P^T$, $\delta_i=1/\lambda_i$ if $\lambda_i \neq 0$, free if $\lambda_i=0$; A^T : $\delta_i=0$ (min norm); $A^T(AA^T)^-$, $(A^T A)^-A^T$ are A^- . $G=A^- \iff AG \text{ idem.}, \rho(AG)=\rho(A)$; AG projects onto $\mathcal{C}(A)$. **Invariance:** $\mathcal{R}(A) \subseteq \mathcal{R}(B)$, $\mathcal{C}(C) \subseteq \mathcal{C}(B) \Rightarrow AB^-C$ free of B^- . Reflexive $A^-AA^-=A^-$; min-norm $(A^-A)^T=A^-A$; LS $(AA^-)^T=AA^-$; A^+ : all three.

[24 Q3] $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, $A_{11} \ r \times r$, $\rho(A)=\rho(A_{11})=r \Rightarrow G = \begin{bmatrix} A_{11}^{-1} & 0 \\ 0 & 0 \end{bmatrix}$ is A^- . $AG = \begin{bmatrix} I_r & 0 \\ A_{21}A_{11}^{-1} & 0 \end{bmatrix}$, $AGA = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{21}A_{11}^{-1}A_{12} \end{bmatrix}$; $[A_{21} \ A_{22}]=M[A_{11} \ A_{12}]$, $M=A_{21}A_{11}^{-1} \Rightarrow A_{22}=A_{21}A_{11}^{-1}A_{12}$.

4. MULTIVARIATE NORMAL

$f=(2\pi)^{-n/2}|\Sigma|^{-1/2}e^{-\frac{1}{2}(y-\mu)^T\Sigma^{-1}(y-\mu)}$; MGF $e^{t^T\mu+\frac{1}{2}t^T\Sigma t}$; $\mathbf{A}\mathbf{Y} \sim N(\mathbf{A}\mu, \mathbf{A}\Sigma\mathbf{A}^T)$. $\mathbf{Y}_1|\mathbf{Y}_2 \sim N(\mu_1+\Sigma_{12}\Sigma_{22}^{-1}(\mathbf{Y}_2-\mu_2), \Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$.

$$\mathbf{A}\mathbf{Y} \perp\!\!\!\perp \mathbf{B}\mathbf{Y} \iff \mathbf{A}\Sigma\mathbf{B}^T=0; \mathbf{Y}^T\mathbf{A}\mathbf{Y} \perp\!\!\!\perp \mathbf{B}\mathbf{Y} \iff \mathbf{B}\Sigma\mathbf{A}=0$$

Moments [HW1 Q1] (A symm., $\mathbf{Y} \sim N_n(\mu, \Sigma)$)

$$\mathbb{E}(\mathbf{Y}^T\mathbf{A}\mathbf{Y}) = \text{tr}(\mathbf{A}\Sigma) + \mu^T\mathbf{A}\mu$$

$$\text{Cov}(\mathbf{Y}, \mathbf{Y}^T\mathbf{A}\mathbf{Y}) = 2\Sigma\mathbf{A}\mu$$

$$\text{Var}(\mathbf{Y}^T\mathbf{A}\mathbf{Y}) = 2\text{tr}((\mathbf{A}\Sigma)^2) + 4\mu^T\mathbf{A}\Sigma\mathbf{A}\mu$$

Distributions: χ_k^2 MGF $(1-2t)^{-k/2}$, $t < \frac{1}{2}$; $\chi_2^2 \sim \text{Exp}(\frac{1}{2})$; $\chi_k^2(\delta)$ MGF $(1-2t)^{-k/2}e^{2t\delta/(1-2t)}$, $\mathbb{E}=k+2\delta$, $\text{Var}=2k+8\delta$. $F_{a,b}=\frac{\chi_a^2/a}{\chi_b^2/b}$, $\mathbb{E}=\frac{b}{b-2}$; $t_\nu=\frac{N(0,1)}{\sqrt{\chi_\nu^2/\nu}}$; $t_\nu^2=F_{1,\nu}$; $\chi_a^2 \perp\!\!\!\perp \chi_b^2 \Rightarrow \frac{A}{A+B} \sim \text{Beta}(\frac{a}{2}, \frac{b}{2})$.

$$\mathbb{E}e^{t^T\mathbf{Y}^T\mathbf{A}\mathbf{Y}} = |\det(I-2tA)|^{-1/2}e^{-\frac{1}{2}t^T\mu+\frac{1}{2}t^T(I-2tA)^{-1}\mu}$$

KEY THM $\mathbf{Y} \sim N_n(\mu, I)$, A symm.:

$$\mathbf{Y}^T\mathbf{A}\mathbf{Y} \sim \chi_{\rho(A)}^2(\tfrac{1}{2}\mu^T\mathbf{A}\mu) \iff A^2 = A$$

central $\iff \mathbf{A}\mu=0$. Pf: $\prod_{j \leq r}(1-2t\lambda_j)^{-1/2}=(1-2t)^{-r/2} \forall t \iff \lambda_j=1$. $\mathbf{Y} \sim N(\mathbf{X}\beta, \sigma^2 I) \Rightarrow \text{SSE}/\sigma^2 \sim \chi_{n-\rho(\mathbf{X})}^2$, $\text{SSE} \perp\!\!\!\perp \hat{\beta}$.

FISHER-COCHRAN $\mathbf{Y} \sim N_n(\mu, I)$, A_j symm. nnd, $\mathbf{Y}^T\mathbf{Y}=\sum_j \mathbf{Y}^T A_j \mathbf{Y}$. Then $\mathbf{Y}^T A_j \mathbf{Y} \sim \chi_{\rho(A_j)}^2(\lambda_j/2)$ mutually independent

$$\iff \sum_j \rho(A_j) = n$$

and then $\lambda_j=\mu^T A_j \mu$, $\sum \lambda_j=\mu^T \mu$, $A_j^2=A_j$, $A_j A_k=0$.

5. GAUSS-MARKOV MODEL

$\mathbf{Y}=\mathbf{X}\beta+\epsilon$, $\mathbf{X} \ n \times p$, $\mathbb{E}\epsilon=0$, $\text{Var}\epsilon=\sigma^2 I$.

$$\mathbf{X}^T\mathbf{X}\beta = \mathbf{X}^T\mathbf{Y} \text{ (always solvable), } \hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-}\mathbf{X}^T\mathbf{Y}$$

$\hat{\mu}=\hat{\mathbf{Y}}=\mathbf{P}\mathbf{X}\mathbf{Y}$; $e=(I-\mathbf{P}\mathbf{X})\mathbf{Y}=(I-\mathbf{P}\mathbf{X})\epsilon$; $\text{Var}(e)=\sigma^2(I-\mathbf{P}\mathbf{X})$; $\text{Var}(\hat{\mathbf{Y}})=\sigma^2\mathbf{P}\mathbf{X}$; $e \perp\!\!\!\perp \hat{\beta}$. Full rank: $\hat{\beta}=(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y} \sim N_p(\beta, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1})$; $H=\mathbf{P}\mathbf{X}$, $\sum h_{ii}=p$, $\text{Var}(e_i)=\sigma^2(1-h_{ii})$.

$$\hat{\sigma}^2 = \text{MSE} = \frac{\mathbf{Y}^T(I-\mathbf{P}\mathbf{X})\mathbf{Y}}{n-\rho(\mathbf{X})}, \quad \mathbb{E}(\text{SSE}) = \sigma^2(n-\rho(\mathbf{X}))$$

Identifiable $\iff \mathbf{X}^T\mathbf{X}$ nonsingular $\iff \rho(\mathbf{X})=p$. **ESTIMABLE** ($\exists \ell^T \mathbf{Y}$ unbiased):

$$a^T\beta \text{ est.} \iff a \in \mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T\mathbf{X}) \iff a^T z = 0 \ \forall z \in \mathcal{N}(\mathbf{X})$$

$\mathbb{E}(\ell^T \mathbf{Y})=a^T\beta \ \forall \beta \iff \ell^T \mathbf{X}=a^T \iff a=\mathbf{X}^T \ell$. Estimable $\Rightarrow a^T \hat{\beta}$ unique. **LZF** $c^T \mathbf{Y}$: $\mathbb{E}=0 \ \forall \beta \iff \mathbf{X}^T c=0$. **BLUE crit.** LUE is BLUE $\iff \text{Cov}(\ell^T \mathbf{Y}, c^T \mathbf{Y})=0$ for every LZF.

$$\mathbf{G-M:} \ a^T\beta \text{ est.} \Rightarrow a^T \hat{\beta} \text{ unique BLUE, } \text{Var} = \sigma^2 a^T (\mathbf{X}^T\mathbf{X})^{-} a$$

$a=\mathbf{X}^T\mathbf{X}u \Rightarrow a^T \hat{\beta}=u^T \mathbf{X}^T \mathbf{Y}$; $\mathbf{X}^T h=0 \Rightarrow \text{Cov}=\sigma^2 u^T \mathbf{X}^T h=0$. $\theta=\mathbf{X}^T \mu=\mathbf{X}^T \mathbf{X}\beta$ always estimable, BLUE $\mathbf{X}^T \mathbf{Y}$ [25 Q2b]. **GLS** $\text{Var}\epsilon=\Sigma \succ 0$: $\mathbf{X}^T \Sigma^{-1} \mathbf{X}\beta=\mathbf{X}^T \Sigma^{-1} \mathbf{Y}$, $\hat{\beta}^G=(\mathbf{X}^T \Sigma^{-1} \mathbf{X})^{-1} \mathbf{X}^T \Sigma^{-1} \mathbf{Y}$; $\text{Var}(a^T \hat{\beta}^G)=a^T (\mathbf{X}^T \Sigma^{-1} \mathbf{X})^{-1} a \leq \text{Var}(a^T \hat{\beta}^{LS})=a^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \Sigma \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} a$.

6. TESTING: WALD = F = LRT [25 Q1,Q3]

$\epsilon \sim N_n(0, \sigma^2 I)$, $\rho(\mathbf{X})=p$, $H_0:L^T\beta=\theta_0$, $\rho(L)=r$, $\hat{\theta}=L^T \hat{\beta}$, $V:=L^T (\mathbf{X}^T \mathbf{X})^{-1} L$, $\hat{\theta} \sim N_r(L^T \beta, \sigma^2 V)$.

$$\tilde{\beta} = \hat{\beta} - (\mathbf{X}^T \mathbf{X})^{-1} L V^{-1} (\hat{\theta} - \theta_0)$$

$$\text{SSE}_{red} - \text{SSE}_{full} = (\hat{\theta} - \theta_0)^T V^{-1} (\hat{\theta} - \theta_0) = \|\mathbf{X}\tilde{\beta} - \mathbf{X}\hat{\beta}\|^2$$

$$W = \frac{1}{\hat{\sigma}^2} (\hat{\theta} - \theta_0)^T V^{-1} (\hat{\theta} - \theta_0) \sim r F_{r, n-p}(\lambda/2)$$

$\lambda=\sigma^{-2}(L^T\beta-\theta_0)^T V^{-1}(L^T\beta-\theta_0)$; $\lambda=0$ under H_0 ; power $\uparrow 1$ as $\lambda \uparrow \infty$.

$$F = \frac{(\text{SSE}_{red} - \text{SSE}_{full})/(\text{df}_{red} - \text{df}_{full})}{\text{MSE}_{full}} = \frac{W}{r} \sim F_{r, n-p}$$

$\text{df}_{full}=n-p$, $\text{df}_{red}=n-p+r$; reject if $F > F_{r, n-p}^{1-\alpha}$. $r=1$: $T = \frac{\hat{\theta}-\theta_0}{\hat{\sigma}\sqrt{V}} \sim t_{n-p}$, $F=T^2$.

$$\Lambda = \left(\frac{\text{SSE}_{full}}{\text{SSE}_{red}} \right)^{n/2}, \quad -2 \log \Lambda \xrightarrow{d} \chi_r^2$$

small $\Lambda \iff \text{large } \frac{\text{SSE}_{red}-\text{SSE}_{full}}{\text{SSE}_{full}} \iff \text{large}$

F . **Sub-model** $\mathbf{X}=[\mathbf{Z}:\mathbf{W}]$, $H_0:\delta=0$: $F = \frac{\mathbf{Y}^T \mathbf{P}_{\mathbf{W}} \mathbf{Z} \mathbf{Y} / s}{\mathbf{Y}^T (I-\mathbf{P}_{\mathbf{X}}) \mathbf{Y} / (n-p)}$; $\mathbf{Y}^T \mathbf{Y}=\mathbf{Y}^T \mathbf{P}_{\mathbf{Z}} \mathbf{Y}+\mathbf{Y}^T \mathbf{P}_{\mathbf{W}|\mathbf{Z}} \mathbf{Y}+\mathbf{Y}^T (I-\mathbf{P}_{\mathbf{X}}) \mathbf{Y}$, ranks $q+s+(n-p)=n$.

7. FUNDAMENTAL THEOREMS OF LS

1st $\text{SSE}=\min_{\beta} \|\mathbf{Y}-\mathbf{X}\beta\|^2=\mathbf{Y}^T(I-\mathbf{P}_{\mathbf{X}})\mathbf{Y} \sim \sigma^2 \chi_{n-\rho(\mathbf{X})}^2$.

2nd $\mathcal{C}(L) \subseteq \mathcal{C}(\mathbf{X}^T)$: $\text{SSE} \perp\!\!\!\perp (\text{SSE}_{red}-\text{SSE})$; $\text{SSE}_{red}-\text{SSE} \sim \sigma^2 \chi_m^2(\delta)$, $\delta=0 \iff L^T\beta=\theta_0$.

3rd [25 Q1] $\mathbf{X}=[\mathbf{Z}:\mathbf{W}]$, $\rho(\mathbf{X})=p$, $\rho(\mathbf{Z})=q < p$; $S_1=\min_{\beta \in \mathbb{R}^p} \|\mathbf{Y}-\mathbf{X}\beta\|^2$, $S_0=\min_{\gamma \in \mathbb{R}^q} \|\mathbf{Y}-\mathbf{Z}\gamma\|^2$; if $\mathbb{E}\mathbf{Y} \in \mathcal{C}(\mathbf{Z})$:

$$\frac{S_1}{S_0} \sim \text{Beta}\left(\frac{n-p}{2}, \frac{p-q}{2}\right)$$

$$S_1 \sim \sigma^2 \chi_{n-p}^2 \perp\!\!\!\perp (S_0-S_1) \sim \sigma^2 \chi_{p-q}^2; \frac{S_1}{S_0} = \frac{S_1}{S_1+(S_0-S_1)}.$$

8. PARTIALLING OUT [24 Q4, 25 Q3]

Frisch-Waugh $\mathbf{X}=[\mathbf{Z}:\mathbf{W}]$, $\beta=(\gamma^T, \delta^T)^T$:

$$\hat{\delta} = (\mathbf{W}^T(I-\mathbf{P}_{\mathbf{Z}})\mathbf{W})^{-1} \mathbf{W}^T(I-\mathbf{P}_{\mathbf{Z}})\mathbf{Y}$$

$$\hat{\gamma} = (\mathbf{Z}^T(I-\mathbf{P}_{\mathbf{W}})\mathbf{Z})^{-1} \mathbf{Z}^T(I-\mathbf{P}_{\mathbf{W}})\mathbf{Y}$$

$\mathbf{Z}^T \mathbf{W}=0 \Rightarrow \hat{\gamma}=(\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{Y}$ (separate fits agree).

[24 Q4] $d_j:=\mathbf{X}_{\cdot j}^T(I-\mathbf{P}_{-j})\mathbf{X}_{\cdot j}$:

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\mathbf{X}_{\cdot j}^T(I-\mathbf{P}_{-j})\mathbf{X}_{\cdot j}} = \frac{\sigma^2}{d_j}$$

$$T_j = \frac{\hat{\beta}_j}{\hat{\sigma}/\sqrt{d_j}} = \frac{1}{\hat{\sigma}/\sigma} \left(\frac{\beta_j}{\sigma} \sqrt{d_j} + W_j \right)$$

$W_j \sim N(0, 1) \perp\!\!\!\perp \hat{\sigma}^2$. Collinear $\Rightarrow (I-\mathbf{P}_{-j})\mathbf{X}_{\cdot j} \approx 0 \Rightarrow d_j \approx 0 \Rightarrow$ noncentrality $\approx 0 \Rightarrow$ no power; true 1-regressor model: $d=\|\mathbf{X}_{\cdot j}\|^2$.

$R^2 \ \mathbf{P}_1=\frac{1}{n}\mathbf{1}\mathbf{1}^T$; $\text{SST}=\mathbf{Y}^T(I-\mathbf{P}_1)\mathbf{Y}$, $\text{SSE}=\mathbf{Y}^T(I-\mathbf{P}_{\mathbf{X}})\mathbf{Y}$, $\text{SSR}=\mathbf{Y}^T(\mathbf{P}_{\mathbf{X}}-\mathbf{P}_1)\mathbf{Y}$:

$$R^2 = \frac{\mathbf{Y}^T(\mathbf{P}_{\mathbf{X}} - \mathbf{P}_1)\mathbf{Y}}{\mathbf{Y}^T(I - \mathbf{P}_1)\mathbf{Y}}, \quad R_{full}^2 - R_{sub}^2 = \frac{\mathbf{Y}^T \mathbf{P}_{\mathbf{W}|\mathbf{Z}} \mathbf{Y}}{\text{SST}} \geq 0$$

Partial corr. $\rho_{YX^{(j)} \cdot \text{rest}} = \frac{(X^{(j)})^T(I-H_{(-j)})\mathbf{Y}}{\sqrt{\mathbf{Y}^T(I-H_{(-j)})\mathbf{Y}} \sqrt{(X^{(j)})^T(I-H_{(-j)})X^{(j)}}}$.

ANCOVA [25 Q3] $Y_{ij}=\mu_i+\beta_1 z_{ij}+\beta_2 w_{ij}+\epsilon_{ij}$; $S_{zz}=\sum_i \sum_j (z_{ij}-\bar{z}_{i\cdot})^2$, and $S_{zw}, S_{ww}, S_{zy}, S_{wy}$ likewise; $M=\begin{bmatrix} S_{zz} & S_{zw} \\ S_{zw} & S_{ww} \end{bmatrix}$:

$$\hat{\beta} = M^{-1} \begin{bmatrix} S_{zy} \\ S_{wy} \end{bmatrix}, \quad \text{Var}(\hat{\beta}) = \sigma^2 M^{-1}, \quad \text{df} = n-r-2$$

$$H_0:\beta_2=0: U = \left(S_{ww} - \frac{S_{zw}^2}{S_{zz}} \right) \frac{\hat{\beta}_2^2}{\hat{\sigma}^2} \sim F_{1, n-r-2}$$

$$H_0:\beta_1=\beta_2=0: V = \frac{1}{\hat{\sigma}^2} \hat{\beta}^T M \hat{\beta}, \quad V/2 \sim F_{2, n-r-2}$$

9. SIMPLE LINEAR REGRESSION

$S_x^2 = \sum (x_i - \bar{x})^2$ (no division), $S_{xY} = \sum (x_i - \bar{x})(Y_i - \bar{Y})$, $S_Y^2 = \sum (Y_i - \bar{Y})^2$.

$$\hat{\beta}_1 = \frac{S_{xY}}{S_x^2} = r_{xY} \frac{s_Y}{s_x} = \sum_i c_i Y_i, \quad c_i = \frac{x_i - \bar{x}}{S_x^2}; \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}, \quad (\mathbf{X}^T \mathbf{X})^{-1} = \frac{1}{n S_x^2} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix}.$$

$$\text{Var} \hat{\beta}_1 = \frac{\sigma^2}{S_x^2}, \quad \text{Var} \hat{\beta}_0 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_x^2} \right), \quad \text{Cov} = \frac{-\sigma^2 \bar{x}}{S_x^2}$$

Normal eqns $\equiv \sum e_i = 0, \sum e_i x_i = 0; \bar{Y} = \bar{Y}; \text{Cov}(\bar{Y}, \hat{\beta}_1) = 0. \hat{\sigma}^2 = \text{SSE}/(n-2); \hat{\sigma}_{ML}^2 = \frac{n-2}{n} \hat{\sigma}^2$ (biased); SST=SSR+SSE; SSR= $\hat{\beta}_1^2 S_x^2$; $R^2 = r_{xY}^2$. No intercept: $\hat{\beta}_1 = \frac{\sum x_i Y_i}{\sum x_i^2}, \text{Var} = \frac{\sigma^2}{\sum x_i^2}, \text{df } n-1.$

$$\hat{Y}_h = \bar{Y} + \hat{\beta}_1 (x_h - \bar{x}), \quad \text{Var}(\hat{Y}_h) = \sigma^2 \left(\frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2} \right)$$

$t^* = t_{n-2}^{1-\alpha/2}: \hat{\beta}_j \pm t^* s(\hat{\beta}_j); \text{mean at } x_h: \hat{Y}_h \pm t^* \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2}};$

$$\text{PI: } \hat{Y}_h \pm t^* \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_x^2}}$$

Ellipse: $(\beta - \hat{\beta})^T (\mathbf{X}^T \mathbf{X}) (\beta - \hat{\beta}) \leq 2 \hat{\sigma}^2 F_{2, n-2}^{1-\alpha}$; Bonferroni rectangle uses $t_{n-2}^{1-\alpha/4}$; $\mathbb{P}(A_0 \cup A_1) \leq \alpha_0 + \alpha_1$.

Add/drop obs. (HW1 Q3) $\hat{\beta}_1 = \hat{\beta}_1^{(-n)} + \frac{x_n - \bar{x}}{S_x^2} e_n^{(-n)};$

$$\hat{\beta}^{(n+1)} - \hat{\beta}^{(n)} = c_{n+1} (Y_{n+1} - \hat{Y}_{n+1}^{(n)}) (\mathbf{X}^T \mathbf{X})^{-1} x_{n+1}$$

$$c_{n+1} = (1 + x_{n+1}^T (\mathbf{X}^T \mathbf{X})^{-1} x_{n+1})^{-1}; Y_{n+1} - \hat{Y}_{n+1}^{(n)} = (Y_{n+1} - \hat{Y}_{n+1}^{(n+1)}) / c_{n+1}.$$

10. ONE-WAY ANOVA [G — NUMERICAL]

$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij} = \mu_i + \varepsilon_{ij}, i \leq r, j \leq n_i, n = \sum n_i; \rho(\mathbf{X}) = r$ (not $r+1$); $\mathcal{N}(\mathbf{X}) = \langle (1, -1, \dots, -1)^T \rangle$. Constraint $\sum w_i \alpha_i = 0, \sum w_i = 1$; natural $w_i = n_i/n$:

$$\sum_i n_i \alpha_i = 0$$

Estimable: $\mu_i; \alpha_i - \alpha_{i'}; \sum c_i \alpha_i$ with $\sum c_i = 0$. Not: μ or α_i alone.

$$\hat{\mu}_i = \bar{Y}_{i..}, \quad \hat{\mu} = \bar{Y}.., \quad \hat{\alpha}_i = \bar{Y}_{i..} - \bar{Y}..$$

SSTO = $\sum \sum (Y_{ij} - \bar{Y}..)^2$; SSTR = $\sum_i n_i (\bar{Y}_{i..} - \bar{Y}..)^2$ (BG); SSE = $\sum \sum (Y_{ij} - \bar{Y}_{i..})^2$ (WG).

COMPUTE $T_i = \sum_j Y_{ij}, G = \sum_i T_i, \text{CF} = G^2/n$:

$$\text{SSTO} = \sum_i \sum_j Y_{ij}^2 - \text{CF}$$

$$\text{SSTR} = \sum_i T_i^2/n_i - \text{CF}$$

$$\text{SSE} = \text{SSTO} - \text{SSTR} = \sum_i (n_i - 1) s_i^2$$

Source	d.f.	SS	MS	E(MS)
Treatments	$r-1$	SSTR	MSTR	$\sigma^2 + \frac{\sum n_i (\mu_i - \mu)^2}{r-1}$
Error	$n-r$	SSE	MSE	σ^2
Total	$n-1$	SSTO		

$H_0: \alpha_1 = \dots = \alpha_r = 0$:

$$F = \frac{\text{MSTR}}{\text{MSE}} = \frac{\text{SSTR}/(r-1)}{\text{SSE}/(n-r)} \sim F_{r-1, n-r}$$

reject if $F > F_{r-1, n-r}^{1-\alpha}$; under H_1 noncentral with $\lambda = \sum n_i \alpha_i^2 / \sigma^2$;
 $\text{MSE} = \frac{\sum_i (n_i - 1) s_i^2}{n-r}$ (pooled). Intervals ($t^* = t_{n-r}^{1-\alpha/2}$, all use MSE):

$$\mu_i: \bar{Y}_{i..} \pm t^* \sqrt{\text{MSE}/n_i}$$

$$\mu_i - \mu_{i'}: \bar{Y}_{i..} - \bar{Y}_{i'..} \pm t^* \sqrt{\text{MSE} \left(\frac{1}{n_i} + \frac{1}{n_{i'}} \right)}$$

Contrast $\theta = \sum c_i \mu_i, \sum c_i = 0$: BLUE $\hat{\theta} = \sum c_i \bar{Y}_{i..}, \text{Var} = \sigma^2 \sum \frac{c_i^2}{n_i}$:

$$\sum_i c_i \bar{Y}_{i..} \pm t^* \sqrt{\text{MSE}} \sqrt{\sum_i c_i^2 / n_i}$$

$r=2: T_{12} = \frac{\bar{Y}_{1..} - \bar{Y}_{2..}}{\hat{\sigma} \sqrt{1/n_1 + 1/n_2}} \sim t_{n-2}, F = T_{12}^2$; noncentrality $\frac{\mu_1 - \mu_2}{\sigma \sqrt{1/n_1 + 1/n_2}}.$

Checks $\sum T_i = G$; SSTR+SSE=SSTO; $(r-1) + (n-r) = n-1$; $\sum n_i \hat{\alpha}_i = 0$; all SS ≥ 0 ; MSE to 4 d.p.

11. WORKED DESIGNS [24 Q2, 25 Q2]

[25 Q2] $Y_{ijk} = \alpha_i + \beta_j + \varepsilon_{ijk}, i, j \in \{1, 2\}, k \leq m$; rows $(1, 0, 1, 0), (1, 0, 0, 1), (0, 1, 1, 0), (0, 1, 0, 1), \rho(\mathbf{X})=3$:

$$\mathcal{N}(\mathbf{X}) = \langle (1, 1, -1, -1)^T \rangle \Rightarrow \text{est.} \iff a_1 + a_2 = a_3 + a_4$$

$\hat{\mu}_{ij} = \bar{Y}_{i..} + \bar{Y}_{.j.} - \bar{Y}...; \mathbf{X}^T \mathbf{Y} = (T_{1..}, T_{2..}, T_{.1.}, T_{.2.})^T.$

- $\alpha_1: a^T z = 1 \neq 0 \Rightarrow$ not estimable.
- $\alpha_1 - \alpha_2$: BLUE $\bar{Y}_{1..} - \bar{Y}_{2..}, \text{Var} = \sigma^2/m.$
- $\alpha_1 + \alpha_2 + \beta_1 + \beta_2 = \mu_{11} + \mu_{22}$: BLUE $2\bar{Y}..., \text{Var} = \sigma^2/m.$
- $\alpha_1 - \alpha_2 + \beta_1 - \beta_2 = \mu_{11} - \mu_{22}$: BLUE $(\bar{Y}_{1..} - \bar{Y}_{2..}) + (\bar{Y}_{.1.} - \bar{Y}_{.2.}), \text{Var} = 2\sigma^2/m.$

[24 Q2] $Y_{1j} = \beta_1, Y_{2j} = \beta_1 + \beta_2, Y_{3j} = \beta_2 + \beta_3, Y_{4j} = \beta_3 + \beta_4$; rows $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix},$

$\det = 1 \Rightarrow$ full rank, all estimable. $\hat{\beta}_1 = \bar{Y}_{1..}, \hat{\beta}_2 = \bar{Y}_{2..} - \bar{Y}_{1..}, \hat{\beta}_3 = \bar{Y}_{3..} - \bar{Y}_{2..} + \bar{Y}_{1..}, \hat{\beta}_4 = \bar{Y}_{4..} - \bar{Y}_{3..} + \bar{Y}_{2..} - \bar{Y}_{1..}, \text{Var} = \frac{\sigma^2}{m}, \frac{2\sigma^2}{m}, \frac{3\sigma^2}{m}, \frac{4\sigma^2}{m}.$
 $H_0: \beta_1 = \beta_2: \hat{\beta}_1 - \hat{\beta}_2 = 2\bar{Y}_{1..} - \bar{Y}_{2..}, \text{Var} = 5\sigma^2/m, T = \frac{2\bar{Y}_{1..} - \bar{Y}_{2..}}{\hat{\sigma} \sqrt{5/m}} \sim t_{4m-4}.$

12. QUANTILES

ν	.95	t_ν .975	.995	1	2	$F_{d_{1,\nu}}^{0.95}$ 3	4	5
4	2.132	2.776	4.604	7.71	6.94	6.59	6.39	6.26
5	2.015	2.571	4.032	6.61	5.79	5.41	5.19	5.05
6	1.943	2.447	3.707	5.99	5.14	4.76	4.53	4.39
8	1.860	2.306	3.355	5.32	4.46	4.07	3.84	3.69
9	1.833	2.262	3.250	5.12	4.26	3.86	3.63	3.48
10	1.812	2.228	3.169	4.96	4.10	3.71	3.48	3.33
12	1.782	2.179	3.055	4.75	3.89	3.49	3.26	3.11
15	1.753	2.131	2.947	4.54	3.68	3.29	3.06	2.90
16	1.746	2.120	2.921	4.49	3.63	3.24	3.01	2.85
18	1.734	2.101	2.878	4.41	3.55	3.16	2.93	2.77
20	1.725	2.086	2.845	4.35	3.49	3.10	2.87	2.71
21	1.721	2.080	2.831	4.32	3.47	3.07	2.84	2.68
24	1.711	2.064	2.797	4.26	3.40	3.01	2.78	2.62
27	1.703	2.052	2.771	4.21	3.35	2.96	2.73	2.57
30	1.697	2.042	2.750	4.17	3.32	2.92	2.69	2.53
40	1.684	2.021	2.704	4.08	3.23	2.84	2.61	2.45
60	1.671	2.000	2.660	4.00	3.15	2.76	2.53	2.37

$z_{.95} = 1.645, z_{.975} = 1.960, z_{.995} = 2.576; t_\nu^2 = F_{1,\nu}; F_{a,b}^{1-\alpha} = 1/F_{b,a}^\alpha.$

13. EXTRAS

ANOVA completed square [G] under $\sum n_i \alpha_i = 0$:

$$G(\mu, \alpha) = \text{SSE} + n(\mu - \hat{\mu})^2 + \sum_i n_i (\alpha_i - \hat{\alpha}_i)^2$$

$\Rightarrow \hat{\mu} = \bar{Y}..., \hat{\alpha}_i = \bar{Y}_{i..} - \bar{Y}..$ and SSTO = SSE + $\sum n_i \hat{\alpha}_i^2$. Geometry: $(I - \mathbf{P}_1) \mathbf{Y} = (I - \mathbf{P}_X) \mathbf{Y} + (\mathbf{P}_X - \mathbf{P}_1) \mathbf{Y}$, orthogonal.

$$\text{Var}(\hat{\alpha}_i) = \sigma^2 \left(\frac{1}{n_i} - \frac{1}{n} \right), \quad \text{Cov}(\bar{Y}_{i..}, \bar{Y}_{j..}) = \frac{\sigma^2}{n}$$

Multiple regression tests $T_j = \frac{\hat{\beta}_j}{\hat{\sigma} \sqrt{[(\mathbf{X}^T \mathbf{X})^{-1}]_{jj}}} \sim t_{n-p}$; overall

$$F = \frac{\text{SSR}/(p-1)}{\text{SSE}/(n-p)} \sim F_{p-1, n-p}; R_{adj}^2 = 1 - \frac{\text{SSE}/(n-p)}{\text{SST}/(n-1)}.$$

Weighted LS $\hat{\beta} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{Y}$ ($\mathbf{W} = \Sigma^{-1}$ gives GLS/BBLUE).

Balanced two-way, $a \times b, n$ reps $Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon$:
SSA = $bn \sum_i (\bar{Y}_{i..} - \bar{Y}..)^2$ (df $a-1$); SSB = $an \sum_j (\bar{Y}_{.j.} - \bar{Y}..)^2$ (df $b-1$); SSAB = $n \sum_i \sum_j (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}..)^2$ (df $(a-1)(b-1)$);
SSE = $\sum \sum \sum (Y_{ijk} - \bar{Y}_{ij.})^2$ (df $ab(n-1)$); total df $abn-1$. Additive model: $\hat{\mu}_{ij} = \bar{Y}_{i..} + \bar{Y}_{.j.} - \bar{Y}...$

Misc. $\mathbb{E}[Y|X]$ minimises $\mathbb{E}[(Y - g(X))^2]$ over all g with $\mathbb{E} g^2 < \infty$. Poisson reg.: $Y|x \sim \text{Poi}(\beta_0 + \beta_1 x), \mathbb{E} = \text{Var} = \beta_0 + \beta_1 x$ (heteroscedastic). Gaussian $\ell = c_n - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{X}\beta\|^2 \Rightarrow \text{MLE} = \text{LSE}$ for β ; $\hat{\sigma}_{ML}^2 = \frac{1}{n} \text{SSE}$. $Z \sim N_n(\mathbf{0}, I), A$ symm. idem. $\Rightarrow Z^T A Z \sim \chi_{\rho(A)}^2$.